



Rigidity of Surfaces with Constant Extrinsic Curvature in Riemannian Product Spaces

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Abstract

The present paper deals with complete surfaces having constant extrinsic curvature in a Riemannian product space $M^2(c) \times \mathbb{R}$, where $M^2(c)$ is a space form with constant sectional curvature $c \in \{-1, 1\}$. In such setting, we find a Simons-type formula for Cheng-Yau's operator which is used to prove that such surfaces are isometric to a cylinder $\mathbb{H}^1 \times \mathbb{R}$, when $c = -1$ or isometric to a slice $\mathbb{S}^2 \times \{t_0\}$ for some $t_0 \in \mathbb{R}$ when $c = 1$. Finally, we extend the result, when $c = -1$, for the Weingarten linear case.

Keywords Product spaces · Omori-Yau maximum principle · Extrinsic curvature · Complete surfaces

1 Introduction

Within the theory of isometric immersions the problem of characterizing surfaces with one of its constant curvatures (or both) constitutes a classical research theme. In this direction, we can highlight the rigidity theorems considering Alexandrov, Liebmann, and Hilbert on constant curvature surfaces' results as the most celebrated outcomes of the theory of submanifolds in the Euclidean space $\mathbb{R}^3$. For generalizations, we quote (Aledo et al. 2005; Montiel and Ros 1991). Nevertheless, we cannot fail to highlight Hopf's theorem (Hopf 1983) on the characterization of totally umbilical spheres as the unique topological spheres of constant mean curvature immersed into a 3-dimensional space form of constant sectional curvature.

On the other hand, the study of surfaces in homogeneous spaces with 4-dimensional isometry group has attracted the attention of many geometers in recent years. We can say that this attention is due to the studies by Abresch, Rosenberg and Meeks which

Dedicated to the memory of Prof. Manofredo P. do Carmo

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have made it possible greater advances in this research area (Abresch and Rosenberg 2004; Meeks and Rosenberg 2004; Rosenberg 2002). Indeed, Abresch and Rosenberg (2004) discovered a holomorphic quadratic differential for surfaces with constant mean curvature in these spaces, and solved Hopf's theorem in relation to these spaces. Moreover, these spaces are also related to Thurston's eight geometries (Thurston and Levy 1997). In particular in the product spaces $\mathbb{H}^2 \times \mathbb{R}$ and $\mathbb{S}^2 \times \mathbb{R}$, Aledo et al. (2007a, b) described the surfaces with constant Gaussian curvature. Later on, Espinar et al. (2009) proved that a complete surface with constant positive extrinsic curvature in $\mathbb{H}^2 \times \mathbb{R}$ and $\mathbb{S}^2 \times \mathbb{R}$ is a rotational sphere.

In Batista (2011), developed a Simons' formula for surfaces immersed with constant mean curvature in the product spaces $M^2(c) \times \mathbb{R}$. Using this formula, and the generalized maximum principle of Omori-Yau, Batista characterized complete surfaces with constant mean curvature in the spaces $\mathbb{H}^2 \times \mathbb{R}$ and $\mathbb{S}^2 \times \mathbb{R}$. Very recently, Barbosa and do Carmo (2018) studied complete surfaces in $\mathbb{H}^2 \times \mathbb{R}$. They proved that such surfaces consist of a cylinder if and only if both the Gaussian curvature and the extrinsic curvature vanish.

Recently, the author in collaboration with de Lima and dos Santos (2018), studied complete linear Weingarten surfaces¹ immersed in a space form $\mathbb{Q}^3(c)$. In such setting, they showed that these surfaces must be either totally umbilical or isometric to a Clifford torus, when $c = 1$; or a circular cylinder, when $c = 0$; or otherwise, a hyperbolic cylinder, when $c = -1$. Here our goal is to extend the technique developed in de Lima and dos Santos (2018) on to the study of complete surfaces with constant extrinsic curvature immersed into the product space $M^2(c) \times \mathbb{R}$, where $M^2(c)$ is a space form with constant sectional curvature $c \in \{-1, 1\}$. As it is well known, a surface in a three-dimension ambient space, besides the Gauss curvature, has, at each point, the extrinsic curvature, which is the product of the principal curvatures. In the Euclidean space, these two curvatures coincide. In this direction, we extend the maximum principle developed in Caminha (2006) in order to characterize such surfaces under certain appropriated restriction on the angle function and the traceless part of the second fundamental form, obtaining natural generalizations of some results found in Batista (2011).

Outline of this paper In this article, we consider surfaces immersed in a product space $M^2(c) \times \mathbb{R}$ with constant extrinsic curvature. Working with Cheng–Yau's operator, we developed a Simons-type formula for these surfaces (cf. Proposition 2.1). Next, we establish a Akutagawa's maximum principle for an arbitrary semi-elliptic operator (cf. Proposition 2.6). Afterwards, under suitable constraints, we apply our maximum principle to prove that a such surface must be a cylinder $\mathbb{H}^1 \times \mathbb{R}$ when $c = -1$ and a slice $\mathbb{S}^2 \times \{t_0\}$, for some $t_0 \in \mathbb{R}$ when $c = 1$ (cf. Theorems 3.1 and 3.4). Also, we extend the Theorem 3.1 for the Weingarten linear case (cf. Theorem 4.4).

2 Mathematical Preliminaries

In this section, we will introduce some basic facts and notations that will appear in the paper. In what follows, we will suppose that all surfaces are orientable and connected.

¹ A surface is said to be Weingarten linear when its first curvatures are linearly related.

2.1 A Simons-Type Formula for Cheng–Yau’s Operator in $M^2(c) \times \mathbb{R}$

Let $M^2(c)$ be a connected Riemannian surface endowed with metric tensor $\langle \cdot, \cdot \rangle_M$ and of constant sectional curvature $c = -1, 1$. The Riemannian product space $M^2(c) \times \mathbb{R}$ is nothing more than the manifold $M^2 \times \mathbb{R}$ provided with the Riemannian metric

$$\langle v, w \rangle_p = \langle (\pi_I)_*v, (\pi_I)_*w \rangle + \langle (\pi_M)_*v, (\pi_M)_*w \rangle_M, \\ p \in M^2 \times \mathbb{R} \text{ and } v, w \in T_p(M \times \mathbb{R})$$

where $\pi_{\mathbb{R}}$ and π_M denote the projections onto the corresponding factor. Associated to the Riemannian product, the vector field

$$\partial_t := (\partial/\partial t)|_{(p,t)}, \quad (p, t) \in \mathfrak{X}(M \times \mathbb{R})$$

is parallel and unitary, that is,

$$\bar{\nabla} \partial_t = 0 \quad \text{and} \quad \langle \partial_t, \partial_t \rangle = 1, \quad (1)$$

where $\bar{\nabla}$ is the Levi–Civita connection of the Riemannian metric of $M^2 \times \mathbb{R}$. Concerning the $M^2(c) \times \mathbb{R}$, let us recall that its curvature tensor satisfies (see Daniel 2007):

$$\bar{R}(X, Y)Z = c(\langle X, Z \rangle Y - \langle Y, Z \rangle X) + c\langle Z, \partial_t \rangle (\langle Y, \partial_t \rangle X - \langle X, \partial_t \rangle Y) \\ + c(\langle Y, Z \rangle \langle X, \partial_t \rangle - \langle X, Z \rangle \langle Y, \partial_t \rangle) \partial_t, \quad (2)$$

where $X, Y, Z \in \mathfrak{X}(M^2 \times \mathbb{R})$ and, $\bar{R}$ is defined by (see O’Neill 1983),

$$\bar{R}(X, Y)Z = \bar{\nabla}_{[X, Y]}Z - [\bar{\nabla}_X, \bar{\nabla}_Y]Z. \quad (3)$$

Let Σ^2 be a connected isometrically immersed surface which we assume to be orientable and oriented by a globally defined unitary normal vector field N . We set A for the second fundamental tensor of the immersion (with respect to the normal direction N). Denoting by ∇ , the Levi–Civita connection of Σ^2 , the Gauss and Weingarten formulas are given by

$$\bar{\nabla}_X Y = \nabla_X Y + \langle AX, Y \rangle N \quad (4)$$

and

$$A(X) = -\bar{\nabla}_X N, \quad (5)$$

for every tangent vector fields $X, Y \in \mathfrak{X}(\Sigma)$.

With respect to surface Σ^2 , we consider two particular functions naturally attached to a such surface Σ^2 , namely, the (vertical) height function $h = (\pi_{\mathbb{R}})|_{\Sigma}$ and the angle function $\langle N, \partial_t \rangle$. In particular, when h is constant, we say that Σ^2 is isometric to a slice $M^2 \times \{t_0\}$ of $M^2(c) \times \mathbb{R}$, for some $t_0 \in \mathbb{R}$. A direct computation shows that the gradient of $\pi_{\mathbb{R}}$ on $M^2(c) \times \mathbb{R}$ is given by

$$\bar{\nabla} \pi_{\mathbb{R}} = \langle \bar{\nabla} \pi_{\mathbb{R}}, \partial_t \rangle \partial_t = \partial_t, \quad (6)$$

so that the gradient of h on Σ^2 is

$$\nabla h = (\bar{\nabla} \pi_{\mathbb{R}})^{\top} = \partial_t^{\top} = \partial_t - \langle N, \partial_t \rangle N, \quad (7)$$

where $(\cdot)^{\top}$ denote the tangential component of a vector field in $\mathfrak{X}(M \times \mathbb{R})$ along Σ^2 . Thus, we get

$$|\nabla h|^2 = |\partial_t^{\top}|^2 = 1 - \langle N, \partial_t \rangle^2. \quad (8)$$

Besides, from (1), Gauss and Weingarten formulas, the Hessian of the height function and the gradient of the angle function satisfy:

$$\nabla_X \nabla h = \langle N, \partial_t \rangle A(X) \quad \text{and} \quad \nabla \langle N, \partial_t \rangle = -A(\nabla h), \quad (9)$$

for every $X \in \mathfrak{X}(\Sigma)$.

Now, we choose an (local) orthonormal frame $\{e_1, e_2, e_3\}$ in $M^2 \times \mathbb{R}$, with dual coframe $\{\omega_1, \omega_2, \omega_3\}$, such that, at each point of Σ^2 , e_1, e_2 are tangent to Σ^2 and $e_3 = N$ is normal to Σ^2 . Then, the second fundamental form A of Σ^2 has the expression

$$A = \sum_{i,j=1}^2 h_{ij} \omega_i \otimes \omega_j$$

with coefficients $h_{ij} = \langle h(e_i), e_j \rangle = \langle A(e_i), e_j \rangle$. The mean curvature H and the squared norm $|A|^2$ of the second fundamental form A are given by

$$H = \frac{1}{2} \text{tr}(A) \quad \text{and} \quad |A|^2 = \sum_{i,j=1}^2 h_{ij}^2$$

and consequently, we have the relation:

$$4H^2 = |A|^2 + 2K_{ext}, \quad (10)$$

where $K_{ext} = \det(A)$ denotes the *extrinsic curvature* of Σ^2 .

Throughout this paper, for simplicity of notation we denote $e_i(h) = \langle e_i, \nabla h \rangle$. The basic equations of the surface are the *Gauss equation*

$$\begin{aligned} R_{ijkl} = & (h_{ik}h_{jl} - h_{il}h_{jk}) + c(\delta_{ik}\delta_{jl} - \delta_{jk}\delta_{il}) + ce_k(h)(\delta_{il}e_j(h) - \delta_{jl}e_i(h)) \\ & + ce_l(h)(\delta_{jk}e_i(h) - \delta_{ik}e_j(h)) \end{aligned} \quad (11)$$

where R_{ijkl} are the components of the curvature tensor of Σ^2 , the *Codazzi equation*

$$h_{ijk} - h_{ikj} = -\bar{R}_{3ijk} = c\langle N, \partial_t \rangle (\delta_{ik}e_j(h) - \delta_{ij}e_k(h)) \quad (12)$$

where h_{ijk} denote the first covariant derivatives of h_{ij} , and satisfy

$$|\nabla A|^2 = \sum_{i,j,k=1}^2 h_{ijk}^2. \quad (13)$$

Finally, the last basic equation is the *Ricci formula*

$$h_{ijkl} - h_{ijlk} = \sum_{m=1}^2 h_{im} R_{mjkl} + \sum_{m=1}^2 h_{jm} R_{mikl} \quad (14)$$

where h_{ijkl} denote the second covariant derivatives of h_{ij} .

From (10) and the Gauss Eq. (11),

$$2K = 2c\langle N, \partial_t \rangle^2 + 4H^2 - |A|^2 = 2c\langle N, \partial_t \rangle^2 + 2K_{ext}. \quad (15)$$

Considering the foregoing, we compute the Laplacian of $|A|^2$. First and foremost, we recall that

$$\frac{1}{2} \Delta |A|^2 = \sum_{i,j,k=1}^2 h_{ijk}^2 + \sum_{i,j,k=1}^2 h_{ij} \Delta h_{ij} \quad (16)$$

where the Laplacian Δh_{ij} of h_{ij} is defined by $\Delta h_{ij} = \sum_k h_{ijkk}$. Thus, from Codazzi Eq. (12) and Ricci formula (14), we deduce that

$$\begin{aligned} \sum_{i,j,k=1}^2 h_{ij} \Delta h_{ij} &= \sum_{i,j,k=1}^2 h_{ij} (\bar{R}_{3ikj,k} + (h_{ikjk} - h_{ikkj}) + \bar{R}_{3kki,j} + h_{kkij}) \\ &= \sum_{i,j,k=1}^2 h_{ij} (\bar{R}_{3ikj,k} + \bar{R}_{3kki,j}) + \sum_{i,j,k,m=1}^2 h_{ij} (h_{mi} R_{mkjk} + h_{mk} R_{mijk}) \\ &\quad + \sum_{i,j=1}^2 h_{ij} (2H)_{ij}. \end{aligned} \quad (17)$$

In relation to these terms, a straightforward computation guarantees that

$$\begin{aligned} \sum_{i,j,k,m=1}^2 h_{ij} (h_{mi} R_{mkjk} + h_{mk} R_{mijk}) &= \sum_{i,k=1}^2 h_{ii}^2 R_{ikik} + \sum_{i,k=1}^2 h_{ii} h_{kk} R_{kii} \\ &= \sum_{i,k=1}^2 (h_{ii}^2 - h_{ii} h_{kk}) R_{ikik} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_{i,k=1}^2 (h_{ii} - h_{kk})^2 R_{ikik} \\
&= (|A|^2 - 2K_{ext})K
\end{aligned} \tag{18}$$

and using the Codazzi Eq. (12) jointly with (8), we also see that

$$\begin{aligned}
\sum_{i,j,k=1}^2 h_{ij} (\bar{R}_{3ikj,k} + \bar{R}_{3kki,j}) &= -2c \langle A^2(\nabla h), \nabla h \rangle + 2cH \langle A(\nabla h), \nabla h \rangle \\
&\quad + 2c \langle N, \partial_t \rangle^2 |A|^2 - 4cH^2 \langle N, \partial_t \rangle^2.
\end{aligned} \tag{19}$$

Therefore, placing the Eqs. (17)–(19) in (16) we obtain

$$\begin{aligned}
\frac{1}{2} \Delta |A|^2 &= |\nabla A|^2 + 2\text{tr}(A \circ \text{Hess } H) + (|A|^2 - 2K_{ext})K - 2c \langle A^2(\nabla h), \nabla h \rangle \\
&\quad + 2cH \langle A(\nabla h), \nabla h \rangle + 2c \langle N, \partial_t \rangle^2 |A|^2 - 4cH^2 \langle N, \partial_t \rangle^2.
\end{aligned} \tag{20}$$

On the other hand, following Cheng and Yau (1977), we introduce an operator $\square$ acting on any smooth function $u : M^2 \rightarrow \mathbb{R}$ by

$$\square(u) = \text{tr}(P \circ \text{Hess } u), \tag{21}$$

where P denotes the first Newton transformation of A , that is, $P : T\Sigma \rightarrow T\Sigma$ is the operator given by

$$P = 2HI - A \tag{22}$$

which also is a self-adjoint linear operator that commutes with A and $\text{tr}(P) = 2H$.

Taking $u = 2H$, from Eq. (10) we obtain the following:

$$\begin{aligned}
\square(2H) &= \text{tr}(P \circ \text{Hess } (2H)) \\
&= 2H \Delta(2H) - 2\text{tr}(A \circ \text{Hess } H) \\
&= \frac{1}{2} \Delta(2H)^2 - 4|\nabla H|^2 - 2\text{tr}(A \circ \text{Hess } H) \\
&= \frac{1}{2} \Delta |A|^2 + \Delta K_{ext} - 4|\nabla H|^2 - 2\text{tr}(A \circ \text{Hess } H).
\end{aligned} \tag{23}$$

Inserting (20) in previous equality, we obtain

$$\begin{aligned}
\square(2H) &= \Delta K_{ext} + |\nabla A|^2 - 4|\nabla H|^2 + (|A|^2 - 2K_{ext})K - 2c \langle A^2(\nabla h), \nabla h \rangle \\
&\quad + 2cH \langle A(\nabla h), \nabla h \rangle + 2c \langle N, \partial_t \rangle^2 |A|^2 - 4cH^2 \langle N, \partial_t \rangle^2.
\end{aligned} \tag{24}$$

For our purposes, it will be more appropriate to deal with the traceless part of A , which is given by $\Phi = A - HI$, with I the identity operator on $T\Sigma$. Then, $\text{tr}(\Phi) = 0$ and

$$|\Phi|^2 = |A|^2 - 2H^2 \geq 0, \tag{25}$$

with equality at $p \in \Sigma^2$ if and only if p is an umbilical point. Now, by the characteristic polynomial of Φ and by identity (25), we have the following equality:

$$-2c\langle A^2(\nabla h), \nabla h \rangle + 2cH\langle A(\nabla h), \nabla h \rangle = -c|\Phi|^2|\nabla h|^2 - 2cH\langle \Phi(\nabla h), \nabla h \rangle \quad (26)$$

and

$$2c\langle N, \partial_t \rangle^2 |A|^2 - 4cH^2\langle N, \partial_t \rangle^2 = 2c\langle N, \partial_t \rangle^2 |\Phi|^2. \quad (27)$$

Besides this, from Eqs. (8) and (15) we obtain:

$$2K - c|\nabla h|^2 + 2c\langle N, \partial_t \rangle^2 = -|\Phi|^2 + 2H^2 + 5c\langle N, \partial_t \rangle^2 - c.$$

Therefore, inserting these three last equations in (24), we obtain the following Simons-type formula for Cheng–Yau’s operator:

Proposition 2.1 *Let Σ^2 be a surface immersed in a product space $M^2(c) \times \mathbb{R}$. Then,*

$$\begin{aligned} \square(2H) = \Delta K_{ext} + |\nabla A|^2 - 4|\nabla H|^2 + |\Phi|^2 \left(-|\Phi|^2 + 2H^2 + 5c\langle N, \partial_t \rangle^2 - c \right) \\ - 2cH\langle \Phi(\nabla h), \nabla h \rangle. \end{aligned}$$

Remark 2.2 It is important to observe that, in the case where the mean curvature H is constant, the Simons-type formula obtained in Proposition 2.1 reduces to Theorem 3.6 of Batista (2011).

2.2 Fundamental Results

We start this section with the Lemma proved in Li (1997) and Simon (1976) for hypersurfaces in a space form having constant scalar curvature. This is still true for surfaces in a product space having constant extrinsic curvature.

Lemma 2.3 *Let Σ^2 be a surface in the product space $M^2(c) \times \mathbb{R}$ with constant extrinsic curvature $K_{ext} \geq 0$. Then*

$$|\nabla A|^2 \geq 4|\nabla H|^2.$$

Proof Since we are supposing that the extrinsic curvature K_{ext} is constant, taking the derivative in (10) we get

$$4H\nabla H = |A|\nabla|A|$$

and consequently

$$16H^2|\nabla H|^2 = |A|^2|\nabla|A||^2.$$

Combining this with Kato’s inequality

$$|\nabla|A||^2 \leq |\nabla A|^2$$

we obtain

$$16H^2|\nabla H|^2 = |A|^2|\nabla|A||^2 \leq |A|^2|\nabla A|^2.$$

Using (10) one more time and that $K_{ext} \geq 0$,

$$|A|^2 |\nabla A|^2 \geq 16H^2 |\nabla H|^2 = 4(|A|^2 + 2K_{ext}) |\nabla H|^2 \geq 4|A|^2 |\nabla H|^2$$

Therefore, we obtain either

$$|A|^2 = 0 \quad \text{and} \quad |\nabla A|^2 = 4|\nabla H|^2 = 0,$$

or

$$|\nabla A|^2 \geq 4|\nabla H|^2$$

as wished. $\square$

Remark 2.4 From Eqs. (10) and (25),

$$|\Phi|^2 = 2(H^2 - K_{ext}). \quad (28)$$

Assume that $K_{ext} \geq 0$. In the case where $K_{ext} > 0$ it follows from (28) that $H(p) \neq 0$ for every $p \in \Sigma^2$. In this case choose on Σ^2 the orientation such that $H > 0$. When $K_{ext} = 0$, we assume that H does not change sign on Σ^2 and we choose on Σ^2 the orientation such that $H \geq 0$.

For the proof of Theorem 3.1, we will also need the following auxiliary result. Its prove follows the same argument as in the proof of Lemma 5 in Alías et al. (2012).

Lemma 2.5 *Let Σ^2 be a surface immersed in the product space $M^2(c) \times \mathbb{R}$ with constant extrinsic curvature. Let μ_- and μ_+ be, respectively, the minimum and the maximum of the eigenvalues of the operator P in (22) at every point $p \in \Sigma^2$. If $K_{ext} \geq 0$, then the operator $\square$ is semi-elliptic, with $\mu_- \geq 0$ and $\mu_+ \leq 4H$.*

Proof Since $K_{ext} \geq 0$, from (10), we have

$$4H^2 = |A|^2 + 2K_{ext} \geq \lambda_i^2,$$

for each principal curvature λ_i of Σ^2 , $i = 1, 2$.

On the other hand, from Remark 2.4,

$$-2H \leq \lambda_i \leq 2H, \quad i = 1, 2,$$

and hence, for each i

$$0 \leq 2H - \lambda_i \leq 4H.$$

However, $\mu_i = 2H - \lambda_i$ are precisely the eigenvalues of P . We conclude that $\mu_- \geq 0$ and $\mu_+ \leq 4H$. $\square$

2.3 A Maximum Principle

Let Σ^n be a Riemannian manifold, and let $L : \mathcal{C}^2(\Sigma) \rightarrow \mathcal{C}^2(\Sigma)$ be a semi-elliptic operator defined by

$$L(u) = \text{tr}(P \circ \text{Hess } u) \quad (29)$$

where $P : T\Sigma \rightarrow T\Sigma$ is a positive semi-definite symmetric tensor. According to Alías et al. (2016), we say that the *Omori–Yau maximum principle* holds on Σ^n for the operator L if, for any function $u \in \mathcal{C}^2(\Sigma)$ with $u^* = \sup_{\Sigma} u < \infty$, there exists a sequence $\{p_k\}_{k \in \mathbb{N}} \subset \Sigma^n$ with the properties

$$u(p_k) > u^* - \frac{1}{k}, \quad |\nabla u(p_k)| < \frac{1}{k} \quad \text{and} \quad Lu(p_k) < \frac{1}{k}$$

for every $k \in \mathbb{N}$. Equivalently, for any function $u \in \mathcal{C}^2(\Sigma)$ with $u_* = \inf_{\Sigma} u > -\infty$, there exists a sequence $\{p_k\}_{k \in \mathbb{N}} \subset \Sigma^n$ with the properties

$$u(p_k) < u_* + \frac{1}{k}, \quad |\nabla u(p_k)| < \frac{1}{k} \quad \text{and} \quad Lu(p_k) > -\frac{1}{k}$$

for every $k \in \mathbb{N}$.

Using this notation we obtain the following extension of Corollary 2.2 due to Caminha (2006) for semi-elliptic operators.

Proposition 2.6 *Let Σ^n be a complete and non-compact Riemannian manifold with sectional curvature bounded from below and let L be a semi-elliptic operator as in (29) with $\sup_{\Sigma} \text{tr}(P) < +\infty$. If $f \in \mathcal{C}^2(\Sigma)$ is a nonnegative function such that $L(f) \geq af^\beta$ for some real numbers $a > 0$ and $\beta > 1$, then $f \equiv 0$.*

For the proof of Proposition 2.6 we will need the following lemma, which is a consequence of Theorem 6.13 of Alías et al. (2016) (see also Alías et al. 2013):

Lemma 2.7 *Let Σ^n be a complete and non-compact Riemannian manifold with sectional curvature bounded from below. Then the Omori–Yau maximum principle holds on Σ^n for any semi-elliptic operator L with $\sup_{\Sigma} \text{tr}(P) < +\infty$.*

Proof of Proposition 2.6 We consider a positive function $\phi : \mathbb{R}_+^* \rightarrow \mathbb{R}_+^*$ to be chosen later, and let $g = \phi \circ f$. Then the gradient and Hessian of g are given by:

$$\nabla g = \phi'(f)\nabla f \quad \text{and} \quad \text{Hess } g = \phi'(f)\text{Hess } f + \phi''(f)\nabla f.$$

From this and from (29),

$$\begin{aligned} L(g) &= \phi'(f)L(f) + \phi''(f)\langle P(\nabla f), \nabla f \rangle \\ &= \phi'(f)L(f) + \frac{\phi''(f)}{\phi'(f)^2}\langle P(\nabla g), \nabla g \rangle, \end{aligned}$$

so that,

$$-\frac{\phi''(f)}{\phi'(f)^2} \langle P(\nabla g), \nabla g \rangle + L(g) = \phi'(f)L(f).$$

Taking $\phi(t) = \frac{1}{(1+t)^\alpha}$, $\alpha > 0$, follows that its two first derivatives satisfy:

$$\phi'(t) = -\alpha\phi(t)^{\frac{\alpha+1}{\alpha}} \quad \text{and} \quad \frac{\phi''(t)}{\phi'(t)^2} = \left(\frac{\alpha+1}{\alpha}\right) \frac{1}{\phi(t)}.$$

Hence

$$\left(\frac{\alpha+1}{\alpha}\right) \langle P(\nabla g), \nabla g \rangle - \phi(f)L(g) = \alpha\phi(f)^{\frac{2\alpha+1}{\alpha}} L(f) \geq a\alpha \frac{f^\beta}{(1+f)^{2\alpha+1}}.$$

If one now takes $\alpha = \frac{\beta-1}{2} > 0$, we arrive at

$$a\alpha \left(\frac{f}{1+f}\right)^\beta \leq -gL(g) + \left(\frac{\alpha+1}{\alpha}\right) \langle P(\nabla g), \nabla g \rangle. \quad (30)$$

On the other hand, since L is semi-elliptic if and only if P is positive semi-definite and $\rho = \sup_\Sigma \text{tr}(P) < +\infty$, we have that

$$\langle P(\nabla g), \nabla g \rangle \leq \rho |\nabla g|^2,$$

and hence from (30)

$$a\alpha \left(\frac{f}{1+f}\right)^\beta \leq -gL(g) + \rho \left(\frac{\alpha+1}{\alpha}\right) |\nabla g|^2. \quad (31)$$

Since g is bounded from below, L is semi-elliptic and $\sup_\Sigma \text{tr}(P) < +\infty$, by the previous theorem, we get a sequence $\{p_k\}_{k \in \mathbb{N}}$ of points in Σ^n such that

$$g(p_k) < g_* + \frac{1}{k}, \quad |\nabla g(p_k)| < \frac{1}{k} \quad \text{and} \quad L(g(p_k)) > -\frac{1}{k}.$$

Therefore, $f(p_k) \rightarrow f^*$, and substituting into (31) we get

$$a\alpha \left(\frac{f(p_k)}{1+f(p_k)}\right)^\beta \leq \frac{1}{k} \left(g_* + \frac{1}{k}\right) + \frac{\rho(\alpha+1)}{k\alpha}.$$

Making $k \rightarrow +\infty$, we get $f^* = 0$, and since $f \geq 0$ this gives $f \equiv 0$. $\square$

3 Surfaces with Constant Extrinsic Curvature

In this section, we present our results of characterization concerning surfaces with constant extrinsic curvature immersed on to a Riemannian product space $\mathbb{S}^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$. For such, we will consider these spaces as hypersurfaces of the four-dimensional Euclidean space $\mathbb{R}^4$ and the four-dimensional Lorentzian space $\mathbb{L}^4 = (\mathbb{R}^4, -dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2)$, respectively:

$$\mathbb{S}^2 \times \mathbb{R} = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{R}^4; x_1^2 + x_2^2 + x_3^2 = 1 \right\}$$

and

$$\mathbb{H}^2 \times \mathbb{R} = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{L}^4; -x_1^2 + x_2^2 + x_3^2 = -1, x_1 > 0 \right\}.$$

From this our first result extends Theorem 4.2 in Batista (2011) to the case where the extrinsic curvature K_{ext} is constant.

Theorem 3.1 *Let Σ^2 be a complete surface in the product space $\mathbb{H}^2 \times \mathbb{R}$ with constant nonnegative extrinsic curvature. Assume that*

$$5 \sup_{\Sigma} \langle N, \partial_t \rangle^2 < 2K_{ext} + 1 \quad \text{and} \quad \langle \Phi(\nabla h), \nabla h \rangle \geq 0. \quad (32)$$

If H is bounded, then Σ^2 is isometric to a cylinder $\mathbb{H}^1 \times \mathbb{R}$.

Proof First of all, since we assume that the extrinsic curvature K_{ext} is constant, we can write the Proposition 2.1 as follows

$$\square(2H) = |\nabla A|^2 - 4|\nabla H|^2 + 2|\Phi|^2 K_{ext} + |\Phi|^2 - 5\langle N, \partial_t \rangle^2 |\Phi|^2 + 2H\langle \Phi(\nabla h), \nabla h \rangle. \quad (33)$$

Taking account that $K_{ext} \geq 0$, from Lemma 2.3,

$$\square(2H) \geq 2|\Phi|^2 K_{ext} + |\Phi|^2 - 5\langle N, \partial_t \rangle^2 |\Phi|^2 + 2H\langle \Phi(\nabla h), \nabla h \rangle.$$

Whereas we suppose $\langle \Phi(\nabla h), \nabla h \rangle \geq 0$, from Remark 2.4, we obtain

$$\square(2H) \geq |\Phi|^2 \left(2K_{ext} - 5\langle N, \partial_t \rangle^2 + 1 \right).$$

Now, considering $d := 2K_{ext} + 1 - 5 \sup_{\Sigma} \langle N, \partial_t \rangle^2 > 0$, we get

$$\square(2H) \geq d|\Phi|^2.$$

On the other hand, Lemma 2.5 guarantees that the operator P is positive semi-definite if $K_{ext} \geq 0$. Thus, from (29) we have that

$$\square(|\Phi|^2) = 2H\square(2H) + 4\langle P(\nabla H), \nabla H \rangle \geq 2H\square(2H).$$

From Remark 2.4 we have $\sqrt{2}H \geq |\Phi|$. Consequently

$$\square(|\Phi|^2) \geq d\sqrt{2}|\Phi|^3. \quad (34)$$

Our purpose is to apply Proposition 2.6 to $|\Phi|^2$. To this end, we need to know that Σ^2 satisfies the conditions of Proposition 2.6. Indeed, by (15) and our hypothesis (32) we observe that

$$\begin{aligned} 2K &= 2K_{ext} - 2\langle N, \partial_t \rangle^2 = 2K_{ext} - 5\langle N, \partial_t \rangle^2 + 3\langle N, \partial_t \rangle^2 \\ &\geq 2K_{ext} - 5 \sup_{\Sigma} \langle N, \partial_t \rangle^2 > -1, \end{aligned}$$

that is, the Gaussian curvature of Σ^2 is bounded from below. Moreover being $\sup_{\Sigma} H < +\infty$ follows that $\sup_{\Sigma} \text{tr}(P) < +\infty$. Thereby, applying Proposition 2.6 with $\beta = 3/2$ we get $|\Phi|^2 = 0$, that is, Σ^2 is totally umbilical. Since K_{ext} is constant, from (28) follows that H is also constant. Hence, using this fact in (33), we must have $\nabla A = 0$, which means that the principal curvatures are constants. Moreover, (34) become equality, implying that $K_{ext} = 0$. In this setting, from the umbilicity, we must have that Σ^2 is a totally geodesic surface (see also Proposition 5.3 of Van der Veken (2008) or Section 4 of Souam and Toubiana (2009) for the same conclusion). Consequently, by (9) we have that the angle function is constant. But, as we suppose that

$$|\langle N, \partial_t \rangle| < \sqrt{0, 2} < 1,$$

one ensures that Σ^2 cannot be a slice. Thus, by using the Codazzi Eq. (12), we see that the angle function is zero. Therefore, we conclude that Σ^2 is isometric to a cylinder $\mathbb{H}^1 \times \mathbb{R}$. $\square$

Remark 3.2 The inequality involving the operator Φ assumed in (32) is the same used in Theorem 4.2 due (Batista 2011).

As is well known, Cheng–Yau’s operator is a divergence type operator if and only if the associated quadratic form P is divergence-free (cf. for instance Li 1997 or Rosenberg 1993). The next Lemma gives us a nice expression of Cheng–Yau’s operator $\square$ in terms of a divergence.

Lemma 3.3 *For a surface Σ^2 immersed in a product space $M^2(c) \times \mathbb{R}$ we have*

$$\begin{aligned} \text{div } \mathcal{V} &= \square(2H) + c|\Phi|^2\langle N, \partial_t \rangle^2 + 2cH^2 \left(2\langle N, \partial_t \rangle^2 - 1 \right) \\ &\quad + c|\nabla h|^2 K - 2cH \langle \Phi(\nabla h), \nabla h \rangle, \end{aligned}$$

where $\mathcal{V} = 2P(\nabla H) + c(\nabla \nabla_h \nabla h)$.

Proof By a standard tensor computation, it is not difficult to see that

$$\operatorname{div}(P(2\nabla H)) = \square(2H) + 2\langle \operatorname{div} P, \nabla H \rangle, \quad (35)$$

where

$$\operatorname{div} P = \sum_{i=1}^2 \nabla P(e_i, e_i)$$

with

$$\nabla P(X, Y) = (\nabla_Y P)X = \nabla_Y(PX) - P(\nabla_Y X)$$

for every $X, Y \in \mathfrak{X}(\Sigma)$. On the other hand, from (22),

$$\nabla P(X, Y) = 2Y(H)X - \nabla A(X, Y), \quad (36)$$

for every $X, Y \in \mathfrak{X}(\Sigma)$. Taking into account the Codazzi Eq. (12), we get

$$\nabla A(X, Y) = \nabla A(Y, X) + c\langle N, \partial_t \rangle (\langle X, \nabla h \rangle Y - \langle Y, \nabla h \rangle X).$$

Therefore, for every $X \in \mathfrak{X}(\Sigma)$ and for every $e_i, i = 1, 2$, we have

$$\begin{aligned} \langle \nabla A(e_i, e_i), X \rangle &= \langle (\nabla_{e_i} A)e_i, X \rangle = \langle e_i, (\nabla_{e_i} A)X \rangle = \langle e_i, \nabla A(X, e_i) \rangle \\ &= \langle e_i, \nabla A(e_i, X) \rangle + c\langle N, \partial_t \rangle \langle e_i, \langle X, \nabla h \rangle e_i - \langle e_i, \nabla h \rangle X \rangle, \end{aligned}$$

which implies

$$\begin{aligned} \langle \operatorname{tr}(\nabla A), X \rangle &= \sum_{i=1}^2 \langle \nabla A(e_i, e_i), X \rangle \\ &= \sum_{i=1}^2 (\langle e_i, (\nabla_X A)e_i \rangle + c\langle N, \partial_t \rangle (\langle X, \nabla h \rangle \langle e_i, e_i \rangle - \langle e_i, \nabla h \rangle \langle X, e_i \rangle)) \\ &= \operatorname{tr}(\nabla_X A) + c\langle N, \partial_t \rangle \langle X, \nabla h \rangle = 2\langle \nabla H, X \rangle + c\langle N, \partial_t \rangle \langle X, \nabla h \rangle, \end{aligned}$$

where it was used that the trace commutes with ∇_X . In other words,

$$\operatorname{tr}(\nabla A) = 2\nabla H + c\langle N, \partial_t \rangle \nabla h.$$

Using this in (36) we can easily see

$$\operatorname{div} P = \operatorname{tr}(\nabla P) = 2\nabla H - 2\nabla H - c\langle N, \partial_t \rangle \nabla h = -c\langle N, \partial_t \rangle \nabla h$$

and hence

$$\operatorname{div}(P(2\nabla H)) = \square(2H) - 2c\langle N, \partial_t \rangle \langle \nabla h, \nabla H \rangle. \quad (37)$$

Now, our goal is to express the last term of the previous equality in terms of a divergence. For such, we consider the vector field $\nabla_{\nabla h} \nabla h$. From (3),

$$\begin{aligned} \operatorname{div}(\nabla_{\nabla h} \nabla h) &= \sum_{i=1}^2 \langle \nabla_{e_i} \nabla_{\nabla h} \nabla h, e_i \rangle \\ &= \sum_{i=1}^2 \left(-R(e_i, \nabla h) \nabla h, e_i \rangle + \langle \nabla_{\nabla h} \nabla_{e_i} \nabla h, e_i \rangle + \langle \nabla_{\nabla_{e_i} \nabla h} \nabla h, e_i \rangle \right) \\ &= |\nabla h|^2 K + \nabla h(\Delta h) + \langle N, \partial_t \rangle^2 |A|^2. \end{aligned} \quad (38)$$

Observe that the term $\nabla h(\Delta h)$ can be computed using (9) as follows

$$\nabla h(\Delta h) = 2\langle N, \partial_t \rangle \langle \nabla h, \nabla H \rangle - 2H \langle A(\nabla h), \nabla h \rangle.$$

Hence, by using this in (38), we get

$$\operatorname{div}(\nabla_{\nabla h} \nabla h) = |\nabla h|^2 K + 2\langle N, \partial_t \rangle \langle \nabla h, \nabla H \rangle - 2H \langle A(\nabla h), \nabla h \rangle + \langle N, \partial_t \rangle^2 |A|^2. \quad (39)$$

Finally, inserting (26) and (27) in (39),

$$\begin{aligned} \operatorname{div}(\nabla_{\nabla h} \nabla h) &= |\nabla h|^2 K - 2H \langle \Phi(\nabla h), \nabla h \rangle + \langle N, \partial_t \rangle^2 |\Phi|^2 + 2H^2 \left(2\langle N, \partial_t \rangle^2 - 1 \right) \\ &\quad + 2\langle N, \partial_t \rangle \langle \nabla h, \nabla H \rangle \end{aligned}$$

which jointly with (37) conclude the desired result. $\square$

An application of Lemma 3.3 is the following characterization of surfaces with zero extrinsic curvature in the product space $\mathbb{S}^2 \times \mathbb{R}$.

Theorem 3.4 *Let Σ^2 be a complete surface in the product space $\mathbb{S}^2 \times \mathbb{R}$ with constant extrinsic curvature. Assume that the extrinsic curvature vanishes and the Gaussian curvature is greater than $2/5$. Then Σ^2 is isometric to a slice $\mathbb{S}^2 \times \{t_0\}$ for some $t_0 \in \mathbb{R}$.*

Proof Taking $c = 1$ and $K_{ext} = 0$ in Proposition 2.1 and Lemma 3.3, we have

$$\operatorname{div} \mathcal{V} = |\nabla A|^2 - 4|\nabla H|^2 + 2|\Phi|^2 \left(4\langle N, \partial_t \rangle^2 - 1 \right) + |\nabla h|^2 K - 4H \langle \Phi(\nabla h), \nabla h \rangle,$$

On the other hand, since Φ is traceless, by the characteristic polynomial of Φ , the following equality $\sqrt{2}|\Phi(\nabla h)| = |\Phi||\nabla h|$ holds. Using this and Cauchy–Schwarz inequality,

$$-4H \langle \Phi(\nabla h), \nabla h \rangle \geq -4H |\langle \Phi(\nabla h), \nabla h \rangle| \geq -4H |\Phi(\nabla h)| |\nabla h| = -2|\Phi|^2 |\nabla h|^2,$$

where we still use $2H^2 = |\Phi|^2$ and Remark 2.4.

Now, we observe that the hypothesis $K_{ext} = 0$ implies the validity of Lemma 2.3. From this, we have

$$\operatorname{div} \mathcal{V} \geq 2|\Phi|^2 \left(4\langle N, \partial_t \rangle^2 - 1 - |\nabla h|^2 \right) + |\nabla h|^2 K.$$

By (8) and (15),

$$\operatorname{div} \mathcal{V} \geq 2|\Phi|^2(5K - 2) + K|\nabla h|^2.$$

Since the Gaussian curvature K is bounded from below by $2/5$, we can apply Bonnet–Myers’ Theorem in order to ensure that Σ^2 is compact. Integrating and using Stokes’ Theorem we find

$$0 = \int_{\Sigma} \operatorname{div} \mathcal{V} d\Sigma \geq \frac{2}{5} \int_{\Sigma} |\nabla h|^2 d\Sigma \geq 0$$

from which we conclude $|\nabla h|^2 = 0$, that is, the function height is constant on Σ^2 . Therefore Σ^2 is isometric to a slice $\mathbb{S}^2 \times \{t_0\}$ of $\mathbb{S}^2 \times \mathbb{R}$ for some $t_0 \in \mathbb{R}$. $\square$

4 A Further Result: Weingarten Linear Surfaces

A natural extension of surfaces having constant extrinsic curvature is the so-called *linear Weingarten*. A surface is said to be Weingarten linear when its first curvatures are linearly related, that is,

$$K_{ext} = aH + b, \quad (40)$$

for constants $a, b \in \mathbb{R}$. Observe that when $a = 0$, (40) reduces to K_{ext} constant.

To study these surfaces, we will consider the following Cheng–Yau’s modified operator

$$L(u) := \square(u) - \frac{a}{2} \Delta(u). \quad (41)$$

In other words, for every $u \in \mathcal{C}^2(\Sigma)$.

$$L(u) = \operatorname{tr}(\tilde{P} \circ \operatorname{Hess} u), \quad (42)$$

with

$$\tilde{P} = P - \frac{a}{2} I. \quad (43)$$

Using the same ideas of the proof of Lemma 2.3, we obtain the extension of Weingarten linear surfaces satisfying $K_{ext} = aH + b$ with $a, b \in \mathbb{R}$.

Lemma 4.1 *Let Σ^2 be a linear Weingarten surface immersed in product space $M^2(c) \times \mathbb{R}$ such that $K_{ext} = aH + b$ with $a^2 + 8b \geq 0$. Then*

$$|\nabla A|^2 \geq 4|\nabla H|^2.$$

Proof Since we suppose that $K_{ext} = aH + b$, we take the derivative in (10) in order to obtain

$$(4H - a)\nabla H = |A|\nabla|A|$$

And hence

$$(4H - a)H^2|\nabla H|^2 = |A|^2|\nabla|A||^2.$$

Combining this with Kato's inequality

$$|\nabla|A||^2 \leq |\nabla A|^2$$

we obtain

$$(4H - a)^2|\nabla H|^2 = |A|^2|\nabla|A||^2 \leq |A|^2|\nabla A|^2. \quad (44)$$

On the other hand, using one more time (10) and also that $K_{ext} = aH + b$,

$$(4H - a)^2 = 16H^2 - 8aH + a^2 = (a^2 + 8b) + 4|A|^2.$$

By this, from (44),

$$|A|^2|\nabla A|^2 \geq (4H - a)^2|\nabla H|^2 = \left((a^2 + 8b) + 4|A|^2\right)|\nabla H|^2 \geq 4|A|^2|\nabla H|^2.$$

Therefore, we obtain either

$$|A|^2 = 0 \quad \text{and} \quad |\nabla A|^2 = 4|\nabla H|^2 = 0,$$

or

$$|\nabla A|^2 \geq 4|\nabla H|^2.$$

□

Remark 4.2 From Eqs. (10) and (25),

$$|\Phi|^2 = 2(H^2 - aH - b). \quad (45)$$

Assume that $b \geq 0$. In the case where $b > 0$, it follows that H has a strict sign. In this case, choose the orientation of Σ^2 so that $H > 0$. When $b = 0$, following Remark 2.4 we assume that H does not change sign on Σ^2 and we choose on Σ^2 the orientation such that $H \geq 0$.

We also have the analogous version of Lemma 2.5.

Lemma 4.3 *Let Σ^2 be a linear Weingarten surface immersed in a product space $M^2(c) \times \mathbb{R}$ such that $K_{ext} = aH + b$. Let $\tilde{\mu}_-$ and $\tilde{\mu}_+$ be, respectively, the minimum and the maximum of the eigenvalues of the operator $\tilde{P}$ defined in (43) at every point $p \in \Sigma^2$. If $b \geq 0$, then the operator L defined in (41) is semi-elliptic, with $\tilde{\mu}_- \geq 0$ and $\tilde{\mu}_+ \leq 4H - a$.*

Proof From (10) and $K_{ext} = aH + b$ with $b \geq 0$, we have

$$4H^2 = |A|^2 + 2aH + 2b \geq \lambda_i^2 + 2aH,$$

for each principal curvature λ_i of Σ^2 , $i = 1, 2$.

On the other hand,

$$\lambda_i^2 \leq 4H^2 - 2aH = \left(2H - \frac{a}{2}\right)^2 - \frac{a^2}{4} \leq \left(2H - \frac{a}{2}\right)^2.$$

We observe that $2H - \frac{a}{2} \geq 0$, indeed, from (10)

$$2H(2H - a) = |A|^2 + 2b \geq 0.$$

From this and Remark 4.2, we get $2H - a \geq 0$. Thus, if $a > 0$ then $2H - \frac{a}{2} > 0$ and, if $a \leq 0$, the assertion is immediate. Thence, for each i , the eigenvalues of $\tilde{P}$ satisfies

$$0 \leq \tilde{\mu}_i = 2H - \frac{a}{2} - \lambda_i \leq 4H - a, \quad i = 1, 2.$$

In particular, $\tilde{\mu}_- \geq 0$ and $\tilde{\mu}_+ \leq 4H - a$. □

With this, we are able to prove the version of linear Weingarten surfaces of Theorem 3.1.

Theorem 4.4 *Let Σ^2 be a complete linear Weingarten surface immersed in the product space $\mathbb{H}^2 \times \mathbb{R}$ such that $K_{ext} = aH + b$ with $b \geq 0$. Assume that*

$$\sup_{\Sigma} \left(\frac{1}{2} |\Phi|^2 + 5 \langle N, \partial_t \rangle^2 \right) < -\frac{a^2}{4} + b + 1 \quad \text{and} \quad \langle \Phi(\nabla h), \nabla h \rangle \geq 0.$$

If the extrinsic curvature is bounded, then Σ^2 is isometric to a cylinder $\mathbb{H}^1 \times \mathbb{R}$.

Proof Since that $K_{ext} = aH + b$, from Proposition 2.1, we write

$$\begin{aligned} \square(2H) &= a\Delta H + |\nabla A|^2 - 4|\nabla H|^2 + |\Phi|^2(2aH + 2b + 1 - 5\langle N, \partial_t \rangle^2) \\ &\quad + 2H\langle \Phi(\nabla h), \nabla h \rangle. \end{aligned} \quad (46)$$

By (41),

$$L(2H) = |\nabla A|^2 - 4|\nabla H|^2 + |\Phi|^2(2aH + 2b + 1 - 5\langle N, \partial_t \rangle^2) + 2H\langle \Phi(\nabla h), \nabla h \rangle.$$

We assume that $b \geq 0$, Lemma 4.1 give us,

$$L(2H) \geq |\Phi|^2(2aH + 2b + 1 - 5\langle N, \partial_t \rangle^2) + 2H\langle \Phi(\nabla h), \nabla h \rangle.$$

and once that $\langle \Phi(\nabla h), \nabla h \rangle \geq 0$,

$$L(2H) \geq |\Phi|^2 \left(2aH + 2b + 1 - 5\langle N, \partial_t \rangle^2 \right). \quad (47)$$

Now, from (45),

$$\left(H - \frac{a}{2} \right)^2 = \frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}.$$

We claim that $H - \frac{a}{2} \geq 0$. Let us consider two cases. When $a \leq 0$ the assertion is immediate. In contrast, when $a > 0$ from (10) we write

$$4H \left(H - \frac{a}{2} \right) = |A|^2 + 2b \geq 0.$$

Thus, from Remark 4.2, we must have $H - \frac{a}{2} \geq 0$ completing the claim. Hence

$$H - \frac{a}{2} = \sqrt{\frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}}. \quad (48)$$

In consideration of the foregoing, by Eq. (15) and completing the square, we arrived at

$$\begin{aligned} 2aH + 2b + 1 - 5\langle N, \partial_t \rangle^2 &= 2a\sqrt{\frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}} - 5\langle N, \partial_t \rangle^2 + a^2 + 2b + 1 \\ &= \left(a + \sqrt{\frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}} \right)^2 - \frac{1}{2}|\Phi|^2 - \frac{a^2}{4} - 5\langle N, \partial_t \rangle^2 \\ &\quad + b + 1 \geq -\frac{1}{2}|\Phi|^2 - 5\langle N, \partial_t \rangle^2 + b - \frac{a^2}{4} + 1. \end{aligned}$$

Hence, we write (47) as

$$L(2H) \geq |\Phi|^2 \left(-\frac{1}{2}|\Phi|^2 - 5\langle N, \partial_t \rangle^2 + b - \frac{a^2}{4} + 1 \right). \quad (49)$$

Consequently, if we take

$$d := b - \frac{a^2}{4} + 1 - \sup_{\Sigma} \left(\frac{1}{2}|\Phi|^2 + 5\langle N, \partial_t \rangle^2 \right) > 0$$

from (49) one gets

$$L(2H) \geq d|\Phi|^2. \quad (50)$$

On the other hand, considering once again that $b \geq 0$, Lemma 4.1 guarantees that the operator $\tilde{P}$ is positive semi-definite. Thereby, from (42) we have that

$$L(|\Phi|^2) = 2HL(2H) + 4\langle \tilde{P}(\nabla H), \nabla H \rangle - aL(2H) \geq 2\left(H - \frac{a}{2}\right)L(2H).$$

In relation to this inequality, from (48) and (50), we obtain the following estimate

$$L(|\Phi|^2) \geq 2d|\Phi|^2\sqrt{\frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}} \geq d\sqrt{2}|\Phi|^3.$$

Whereas that $\sup_{\Sigma} K_{ext} < +\infty$, as Σ^2 is a linear Weingarten surface satisfying $K_{ext} = aH + b$ we must have $\sup_{\Sigma} H < +\infty$. Hence,

$$\sup_{\Sigma} \text{tr}(\tilde{P}) = 2 \sup_{\Sigma} H - a < +\infty.$$

Moreover, as $b \geq 0$, Lemma 4.3 give us the semi-ellipticity of $\tilde{P}$. Now, from Gauss Eq. (11), we see that

$$\begin{aligned} 2K &= -2\langle N, \partial_t \rangle^2 + 2aH + 2b \\ &= -5\langle N, \partial_t \rangle^2 + 3\langle N, \partial_t \rangle^2 + 2a\sqrt{\frac{1}{2}|\Phi|^2 + b + \frac{a^2}{4}} + a^2 + 2b \\ &\geq -\frac{1}{2}|\Phi|^2 - 5\langle N, \partial_t \rangle^2 + b - \frac{a^2}{4} \\ &\geq -\sup_{\Sigma} \left(\frac{1}{2}|\Phi|^2 + 5\langle N, \partial_t \rangle^2 \right) + b - \frac{a^2}{4} > -1, \end{aligned}$$

in other words, the Gaussian curvature is bounded from below. Hence, from Proposition 2.6 we obtain that $|\Phi| = 0$, that is, Σ^2 is totally umbilical. Being $K_{ext} = aH + b$ from (45) follows that H and K_{ext} are constants. At this point, we can reason as in the last part of the proof of Theorem 3.1 to conclude the result. $\square$

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